Light

GCSE Computer Science - Programming Testing and Robustness (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of testing a program?

- ○ **A.** To make the program run faster
- ○ **B.** To check that the program works correctly and handles different inputs appropriately
- ○ **C.** To add more features to the program
- ○ **D.** To reduce the size of the program file

Q2 (1 mark)

What is data validation?

- ○ **A.** Encrypting data to keep it secure
- ○ **B.** Checking that data entered meets specified rules or criteria before it is processed
- ○ **C.** Deleting incorrect data from a database
- ○ **D.** Converting data from one format to another

Q3 (1 mark)

What is normal (typical) test data?

- ○ **A.** Data that a user would typically enter and the program should accept
- ○ **B.** Data that is at the very edge of acceptable values
- ○ **C.** Data that is deliberately wrong to test error handling
- ○ **D.** Data that causes the program to crash

Q4 (1 mark)

What is a syntax error?

- ○ **A.** An error in the logic of the program that causes incorrect output
- ○ **B.** An error where the program runs but produces the wrong results
- ○ **C.** An error that only occurs when specific data is entered

- ○ **D.** An error that breaks the grammatical rules of the programming language, preventing the program from running

Q5 (1 mark)

What is a logic error?

- ○ **A.** An error where the code is grammatically correct but produces incorrect or unexpected results
- ○ **B.** An error that prevents the program from running
- ○ **C.** An error message displayed by the programming environment
- ○ **D.** An error in the computer's processor

Q6 (1 mark)

For a program that accepts values between 1 and 100, which of these would be boundary test data?

- ○ **A.** 50
- ○ **B.** 0, 1, 100, 101
- ○ **C.** -5
- ○ **D.** One hundred

Q7 (1 mark)

What is erroneous test data?

- ○ **A.** Data that the program should accept and process normally
- ○ **B.** Data that is at the boundary of acceptable values
- ○ **C.** Data that is incorrect, invalid, or of the wrong type and should be rejected by the program
- ○ **D.** Data that has been encrypted

Q8 (1 mark)

What is the purpose of authentication in a program?

- ○ **A.** To verify that a user is who they claim to be before granting access
- ○ **B.** To make the program run faster
- ○ **C.** To validate that entered data is in the correct format
- ○ **D.** To compress files to save storage space

Q9 (1 mark)

Which validation check would ensure that a user enters an age between 18 and 65?

- ○ **A.** Length check
- ○ **B.** Format check
- ○ **C.** Presence check
- ○ **D.** Range check

Q10 (1 mark)

What does a presence check validate?

- ○ **A.** That the data is within an acceptable range
- ○ **B.** That something has been entered and the field is not left empty
- ○ **C.** That the data matches a specific pattern
- ○ **D.** That the password is correct

Q11 (1 mark)

A password must be at least 8 characters long. Which validation check should be used?

- ○ **A.** Length check
- ○ **B.** Type check
- ○ **C.** Range check
- ○ **D.** Lookup check

Q12 (1 mark)

Why is it important to test a program with different types of test data?

- ○ **A.** To make the program more colourful
- ○ **B.** To change the programming language
- ○ **C.** To increase the number of lines of code

○ **D.** To ensure the program handles expected inputs, edge cases, and invalid inputs correctly

Q13 (2 marks)

Identify the type of error in this code:

```
total = 10
print(totl)
```

○ **A.** Logic error - the calculation is incorrect

○ **B.** No error - the code will run correctly

○ **C.** Syntax error - the variable name is misspelled

○ **D.** Runtime error - division by zero

Q14 (2 marks)

A program should calculate the average of three numbers. Identify the type of error:

```
num1 = 10
num2 = 20
num3 = 30
average = num1 + num2 + num3 / 3
print(average)
```

○ **A.** Syntax error - missing brackets

○ **B.** Logic error - incorrect order of operations gives wrong result

○ **C.** No error - the code works correctly

- **D.** Runtime error - the program will crash

Q15 (2 marks)

A program accepts exam scores between 0 and 100 (inclusive). Which set of test data best covers normal, boundary, and erroneous cases?

- **A.** -1, 0, 50, 100, 101, "fifty"
- **B.** 50, 75, 90
- **C.** 0, 100
- **D.** 1, 2, 3, 4, 5

Q16 (2 marks)

What is output when the user enters 15?

```
age = INPUT("Enter age: ")
IF age >= 18 THEN
    OUTPUT "Adult"
ELSE
    OUTPUT "Minor"
ENDIF
```

- **A.** Adult
- **B.** Error
- **C.** 15
- **D.** Minor

Q17 (2 marks)

What is output when username "admin" and password "secret" are entered?

```
storedUser = "admin"
storedPass = "secret123"

inputUser = INPUT("Username: ")
inputPass = INPUT("Password: ")

IF inputUser = storedUser AND inputPass = storedPass THEN
    OUTPUT "Access granted"
ELSE
    OUTPUT "Access denied"
ENDIF
```

- ○ **A.** Access granted
- ○ **B.** Access denied
- ○ **C.** admin
- ○ **D.** secret

Q18 (2 marks)

This code validates a percentage score. What is output when 105 is entered?

```
score = INPUT("Enter score: ")
IF score >= 0 AND score <= 100 THEN
    OUTPUT "Valid score"
ELSE
    OUTPUT "Invalid score"
ENDIF
```

- ○ **A.** Valid score

- B. 100
- C. 105
- D. Invalid score

Q19 (2 marks)

This code should count from 1 to 5 but has a logic error. What is wrong?

```
count = 0
WHILE count < 5
    OUTPUT count
    count = count + 1
ENDWHILE
```

- A. The loop never runs
- B. The loop runs forever (infinite loop)
- C. The loop outputs 0, 1, 2, 3, 4 instead of 1, 2, 3, 4, 5
- D. There is a syntax error

Q20 (2 marks)

What is output when "Hi" is entered?

```
password = INPUT("Enter password: ")
IF LENGTH(password) >= 8 THEN
    OUTPUT "Password accepted"
ELSE
    OUTPUT "Password too short"
ENDIF
```

- A. Password too short
- B. Password accepted
- C. Hi
- D. 2

Q21 (2 marks)

Identify the error in this code:

```
IF score > 50
    OUTPUT "Pass"
ELSE
    OUTPUT "Fail"
ENDIF
```

- A. Logic error - wrong comparison operator
- B. Runtime error - score is not defined
- C. No error - the code is correct
- D. Syntax error - missing THEN after the condition

Q22 (2 marks)

A program accepts month numbers from 1 to 12. A student chooses to test with: 6, 1, 12, 0, 13. Which justification is correct?

- A. 6 is erroneous data because it is in the middle
- B. 0 and 13 are boundary data because they are just outside the valid range

- C. 1 and 12 are normal data because they are within the range
- D. All five values are erroneous data

Q23 (2 marks)

What is output when the user presses Enter without typing anything?

```
name = INPUT("Enter name: ")
IF name = "" THEN
    OUTPUT "Name required"
ELSE
    OUTPUT "Hello, " + name
ENDIF
```

- A. Hello,
- B. Error
- C. Hello, name
- D. Name required

Q24 (3 marks)

A program validates a PIN code. Analyse the code and determine what is output when "12A4" is entered.

```
pin = INPUT("Enter 4-digit PIN: ")
valid = True

IF LENGTH(pin) != 4 THEN
    valid = False
    OUTPUT "PIN must be 4 digits"
ENDIF

FOR i = 0 TO LENGTH(pin) - 1
```

```
    IF pin[i] < "0" OR pin[i] > "9" THEN
        valid = False
        OUTPUT "PIN must contain only digits"
    ENDIF
ENDFOR

IF valid = True THEN
    OUTPUT "PIN accepted"
ENDIF
```

- A. PIN accepted
- B. PIN must be 4 digits
- C. PIN must contain only digits
- D. PIN must be 4 digits and PIN must contain only digits

Q25 (3 marks)

This authentication system allows 3 attempts. What is output if the user enters "wrong" three times?

```
correctPassword = "secret"
attempts = 0
loggedIn = False

WHILE attempts < 3 AND loggedIn = False
    password = INPUT("Enter password: ")
    attempts = attempts + 1
    IF password = correctPassword THEN
        loggedIn = True
        OUTPUT "Welcome!"
    ELSE
        OUTPUT "Incorrect password"
    ENDIF
ENDWHILE

IF loggedIn = False THEN
```

```
    OUTPUT "Account locked"
ENDIF
```

- A. Incorrect password (three times), then Account locked
- B. Welcome!
- C. Incorrect password (once only)
- D. Account locked (only)

Q26 (3 marks)

This code should find the largest of three numbers but contains a logic error. Identify the issue.

```
a = 15
b = 28
c = 22

IF a > b AND a > c THEN
    largest = a
ELSE IF b > c THEN
    largest = b
ELSE
    largest = c
ENDIF

OUTPUT largest
```

- A. The code has a syntax error and will not run
- B. The code works correctly with no error
- C. The ELSE IF condition should be b > a AND b > c to properly check if b is largest

- ○ **D.** The variables are declared incorrectly

Q27 (3 marks)

A program validates UK postcodes. Valid postcodes have 6-8 characters and must start with a letter. Design appropriate test data.

- ○ **A.** Normal: "SW1A 1AA", Boundary: "A1 1AA" (6 chars), "AB12 3CDE" (9 chars), Erroneous: "123456", ""
- ○ **B.** Only test "SW1A 1AA" as it is a real postcode
- ○ **C.** Test every possible postcode combination
- ○ **D.** Normal: 7, Boundary: 6 and 8, Erroneous: 5 and 9

Q28 (3 marks)

Examine this code that should calculate a 10% discount. Identify all errors.

```
price = INPUT("Enter price ")
discount = price * 0.1
final = price - discount
PRINT final price
```

- ○ **A.** One syntax error only: the last line has incorrect syntax
- ○ **B.** One logic error only: the discount calculation is wrong
- ○ **C.** Two errors: missing colon after INPUT prompt, and PRINT statement has incorrect syntax
- ○ **D.** No errors - the code is correct

Q29 (3 marks)

A program must accept a quantity between 1 and 99. Analyse which validation code is most robust.

- ○ **A.** IF quantity > 0 AND quantity < 100 THEN valid
- ○ **B.** IF quantity != 0 THEN valid
- ○ **C.** IF quantity >= 1 AND quantity <= 99 THEN valid
- ○ **D.** IF quantity < 100 THEN valid

Q30 (3 marks)

A student tests this code with input 5 and expects output 15, but gets 25. Trace through to identify the logic error.

```
number = INPUT("Enter number: ")
result = 0
FOR i = 1 TO number
    result = result + number
ENDFOR
OUTPUT result
```

- ○ **A.** The loop runs too few times
- ○ **B.** The loop should add i to result, not number - it currently adds the input value each iteration instead of the loop counter
- ○ **C.** The result should start at 1, not 0
- ○ **D.** There is no error - 25 is the correct output

Total: **55** marks • Time: **30 min**

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